Mahindra University
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ÉCOLE CENTRALE SCHOOL OF ENGINEERING

Mathematics-1 (MA 1101)
Tutorial Sheet-2
Topics: Integration, Polar Coordinates and Fundamental Theorem of Calculus

*Problems to be worked out in tutorial

†Assignment Problems

1. † Compute the following integral, and draw the graph of f.

$$\int_0^1 f(x)dx \quad \text{where} \quad f(x) = \begin{cases} x & \text{if} \quad 0 \le x \le c \\ c\dfrac{1-x}{1-c} & \text{if} \quad c \le x \le 1 \end{cases}$$

   c is a fixed real number, $0 < c < 1$.

2. Compute the area of the region S between the graphs of f and g over the interval $[a, b]$ specified in each case. Make a sketch of the two graphs and indicate S by shading.

   i) * $f(x) = |x|, g(x) = x^2 - 1, a = -1, b = 1$

   ii) † $f(x) = |x| + |x - 1|, g(x) = 0, a = -1, b = 2$

3. Sketch the graph of f in polar coordinates and compute the area of the radial set of f over the interval specified. Assume that each set is measurable.

   i) Circle tangent to x-axis: $f(\theta) = 4\sin\theta, 0 \le \theta \le \pi$

   ii) * Lazy eight: $f(\theta) = \sqrt{|\cos\theta|}, 0 \le \theta \le 2\pi$

   iii) † Four-leaved rose: $f(\theta) = |\sin 2\theta|, 0 \le \theta \le 2\pi$

4. Compute the volume of the solid generated by revolving the ordinate set of the function f over the interval indicated. Sketch each of the ordinate sets.

   i) * $f(x) = x^2, -1 \le x \le 2$.

   ii) † $f(x) = \sin x + \cos x, 0 \le x \le \pi$.

5. Compute the average $A(f)$ for the given function f over the specified interval.

   i) $f(x) = x^{\frac{1}{2}}, 0 \le x \le 4$.

   ii) † $f(x) = \sin x \cos x, 0 \le x \le \pi/4$.

6. i) * If $f(x) = x^2$ for $0 \le x \le a$, find a number c satisfying $0 < c < a$ such that $f(c)$ is equal to the average of f in $[0, a]$.

   ii) Solve (i) if $f(x) = x^n$, where n is any positive integer.

7. * The graphs of $f(x) = x^2$ and $g(x) = cx^3$, where $c > 0$, intersect at the points $(0, 0)$ and $(\frac{1}{c}, \frac{1}{c^2})$. Find c so that the region which lies between these graphs and over the interval $[0, \frac{1}{c}]$ has area $\frac{2}{3}$.

8. Compute $f(0)$, given that $f(\pi) = 2$ and that $\displaystyle\int_0^{\pi} [f(x) + f''(x)] \sin x \mathrm{d}x = 5$.

9. * Without attempting to evaluate the integral, compute $f'(x)$ if f is defined by the formula $f(x) = \int_{x^3}^{x^2} \frac{t^6}{1+t^4} dt$

10. * Let $f(x) = \displaystyle\int_x^2 \frac{1}{\sqrt{1+t^3}} \mathrm{d}t$. Find $\displaystyle\int_0^2 x f(x) \mathrm{d}x$.

11. i) † Show that $\displaystyle\int_0^{\pi} x f(\sin x) \mathrm{d}x = \frac{\pi}{2} \int_0^{\pi} f(\sin x) \mathrm{d}x$.

    ii) Use part (i) to deduce the formula $\displaystyle\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \mathrm{d}x = \pi \int_0^1 \frac{1}{1+x^2} \mathrm{d}x$.

12. A function g, defined for all positive real numbers, satisfies the following two conditions: $g(1) = 1$ and $g'(x^2) = x^3$ for all $x > 0$. Compute $g(4)$.

13. † Let $f(x) = \int_0^x (1 + t^3)^{-1/2} dt$ if $x \geq 0$. (Do not attempt to evaluate this integral.)

    i) Show that f is strictly increasing on the nonnegative real axis.

    ii) Let g denote the inverse of f. Show that the second derivative of g is proportional to g^2 [that is, $g''(y) = cg^2(y)$ for each y in the domain of g] and find the constant of proportionality.

14. Without attempting to evaluate the following indefinite integrals, find the derivative $f'(x)$ in each case if $f(x)$ is equal to

    i) * $\displaystyle\int_{x^3}^{x^2} (1 + t^2)^{-3} \mathrm{d}t$

    ii) † $\displaystyle\int_1^{\int_1^x t^2 \mathrm{d}t} t \mathrm{d}t$